

CNM-Au8 in Amyotrophic Lateral Sclerosis

The HEALEY ALS Platform Trial

Writing Committee for the HEALEY ALS Platform Trial

IMPORTANCE Bioenergetic failure has been proposed as a driver of amyotrophic lateral sclerosis (ALS). CNM-Au8 is a suspension of gold nanocrystals that catalyzes the conversion of nicotinamide adenine dinucleotide hydride into NAD⁺, resulting in an increase of cellular adenosine triphosphate production.

OBJECTIVE To determine the effects of CNM-Au8 on ALS disease progression.

DESIGN, SETTING, AND PARTICIPANTS CNM-Au8 was tested as a regimen of the HEALEY ALS Platform Trial, a phase 2/3, multicenter, randomized, double-blind platform trial. The study was conducted at 54 sites in the US from July 2020 to March 2022 (final follow-up, March 17, 2022). A total of 161 participants with ALS were randomized to receive CNM-Au8 (n = 120) or regimen-specific placebo (n = 41). Data from 123 concurrently randomized placebo participants in other regimens were combined for analyses.

INTERVENTIONS Eligible participants were randomized in a 3:3:2 ratio to receive CNM-Au8 60 mg daily (n = 61), CNM-Au8 30 mg daily (n = 59), or matching placebo (n = 41) for 24 weeks.

MAIN OUTCOMES AND MEASURES The primary efficacy outcome was change from baseline through week 24 in ALS disease severity measured by a bayesian shared parameter model of function (based on the Revised Amyotrophic Lateral Sclerosis Functional Rating Scale) and survival, which provided an estimate of the rate of disease progression measured by the disease rate ratio (DRR), with a DRR of less than 1 indicating treatment benefit. Secondary end points included a Combined Assessment of Function and Survival using a joint-rank test, rate of decline in slow vital capacity (percent predicted), and survival free of permanent assisted ventilation.

RESULTS Among 161 participants who were randomized within the CNM-Au8 regimen (mean age, 58.4 years; 61 [37.9%] female), 145 (90%) completed the trial. In the primary analysis comparing the combined CNM-Au8 dosage groups vs the combined placebo groups, the primary end point (DRR, 0.97 [95% credible interval, 0.783-1.175]; posterior probability of DRR <1, 0.65) and the 3 secondary end points suggested no benefit or harm of CNM-Au8. In the active (n = 120) vs placebo (n = 163) groups, the most common adverse events were diarrhea (23 [19%] vs 12 [7%]), nausea (17 [14.2%] vs 14 [8.6%]), fatigue (12 [10.8%] vs 30 [18.4%]), and muscular weakness (24 [20%] vs 45 [27.6%]).

CONCLUSIONS AND RELEVANCE No benefit of CNM-Au8 on ALS disease progression was observed at 24 weeks.

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Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease with a phenotypically heterogeneous presentation causing progressive weakness. Many mechanisms have been proposed to drive ALS onset and progression. Metabolic dysregulation from bioenergetic failure may underlie neurodegeneration in multiple cell types of the central nervous system, including neurons, astrocytes, and oligodendrocytes,^{1,2} and in other systems.³ Therefore improving bioenergetics is an attractive target for novel ALS therapeutics.

CNM-Au8 is a concentrated, aqueous suspension of clean-surfaced faceted gold (Au) nanocrystals that catalyze specific intracellular biological reactions. This compound serves as a catalyst for the intracellular conversion of nicotinamide adenine dinucleotide hydride into NAD⁺, increasing cellular energy production. NAD⁺ enrichment by CNM-Au8 has been demonstrated in vitro using primary neuron-glia co-cultures.^{4,5} Studies in ALS models harboring variations in the superoxide dismutase type-1 (*SOD1*) gene, accounting for approximately 2% of individuals with ALS, have highlighted the positive effects of NAD⁺ availability on ALS astrocyte-mediated neurotoxicity in vitro.^{6,7} NAD⁺ has also been shown to be important in improving neuronal metabolism, modulating mitochondrial proteostasis, and improving neurogenesis in the brains of mutant TAR DNA binding protein 43 (*TARDBP*) mice, characterized by cytoplasmic aggregation of TDP-43 protein in neuronal and non-neuronal cells, and *SOD1* mice.^{8,9} NAD⁺ deficits, coupled with mitochondrial dysfunction, have also been demonstrated in differentiated motor neurons derived from human patient-induced pluripotent stem cells, representing both familial and sporadic ALS cases.¹⁰ A phase 2, randomized, double-blind study of CNM-Au8 in a cohort of sporadic ALS participants (n = 45) has been completed, supporting the favorable safety profile of CNM-Au8.^{11,12} Here, the results of the randomized, double-blind, placebo-controlled trial of CNM-Au8 are reported, tested in the HEALEY ALS Platform Trial, a perpetual design platform trial planned to test multiple investigational products in parallel against placebo.¹³

Methods

Trial Design and Oversight

In the HEALEY ALS Platform Trial, a phase 2/3 adaptive platform trial for people living with ALS, each new investigational product is added as a regimen to the master protocol; CNM-Au8 was evaluated in regimen C. Because of the shared protocol and infrastructure, data from placebo participants in other regimens can be shared and included in the analyses. Randomization in regimen C was 3:3:2 (CNM-Au8 60 mg daily, CNM-Au8 30 mg daily, and placebo). Regimen C was conducted at 54 sites across the US from July 2020 to March 2022.¹³ Protocol approval was provided for all trial sites by a central institutional review board, the Massachusetts General Brigham Human Research Committee. The trial was designed by and conducted through the [NEALS network](#), a global collaborative trial network, in collaboration with Clene Nanomedicine. Clene Nanomedicine provided the active drug and pla-

Key Points

Question What is the effect of CNM-Au8, an oral suspension of catalytically active gold nanocrystals, on amyotrophic lateral sclerosis progression over 24 weeks?

Findings In this randomized platform trial testing multiple regimens, a total of 161 participants were randomized to receive CNM-Au8 (n = 120) or regimen-specific placebo (n = 41); analyses included an additional 123 concurrently randomized placebo participants from other regimens. The disease rate ratio (assessing disease progression accounting for survival, with <1 indicating slowing of disease progression on CNM-Au8 relative to placebo) was 0.97, suggesting no significant benefit or harm of CNM-Au8 on disease progression over 24 weeks.

Meaning No significant benefit of CNM-Au8 on amyotrophic lateral sclerosis disease progression was observed over 24 weeks.

cebo and was involved in the trial design, data analysis, and manuscript review. Trial reporting follows the [CONSORT](#) reporting guidelines for randomized clinical trials. Additional details are available in the eMethods in [Supplement 3](#).

Trial Participants

Eligibility criteria for the platform trial included adults with a diagnosis of clinically possible, probable, laboratory-supported probable, or definite ALS defined by the revised El Escorial criteria.

Written informed consent was obtained from all participants. Additional details are available in the eMethods in [Supplement 3](#) and in the trial protocol in [Supplement 1](#). Data on race and ethnicity were collected to help establish racial representation in this trial, explain trial generalizability to external populations, and encourage health equity. Self-reported race and ethnicity data were collected following National Institute of Neurological Disorders and Stroke recommended common data elements using fixed categories.

Trial Interventions and Procedures

Random assignment to a regimen was not blinded to either the participant or investigator, whereas randomization to treatment group within a regimen was blinded to both the participant and investigator. Eligible participants were randomized in a 3:3:2 ratio to receive CNM-Au8 60 mg daily, CNM-Au8 30 mg daily, or matching placebo, with stratification for edaravone and/or riluzole use. Additional details are available in the eMethods in [Supplement 3](#).

Outcomes

The primary end point was change in disease severity from baseline through 24 weeks measured by the Revised Amyotrophic Lateral Sclerosis Functional Rating Scale (ALSFRS-R) total score and survival. A bayesian shared parameter model of function and survival was used, providing an integrated estimate of the increase or decrease in the rate of disease progression on treatment relative to control, or disease rate ratio (DRR [DRR <1 indicates benefit]).¹⁴ The secondary clinical efficacy outcomes included (1) the Combined Assessment of

Function and Survival (CAFS) model,^{15,16} (2) rate of decline in percent predicted normal (PPN) slow vital capacity (SVC), and (3) survival that is free of permanent assisted ventilation (PAV). The ALSFRS-R total score ranges from 0 to 48, with higher values indicating better function. The CAFS is a joint-rank analysis of function and survival wherein higher rankings indicate benefit. SVC measures the maximum volume of air that can be slowly exhaled after maximal inhalation and is converted into PPN using published values,¹⁷ with higher values indicating greater respiratory function. Prespecified exploratory measures included biomarkers such as neurofilament light (NfL) levels,^{18,19} serum creatinine, and urinary p75 extracellular domain. Other exploratory measures included clinical outcome measures (handheld dynamometry, quantitative motor speech, home vital capacity, ALSFRS-R domain scores, the Amyotrophic Lateral Sclerosis Assessment Questionnaire and its domain scores, and the Center for Neurologic Study - Bulbar Function Scale) and time-to-event analyses (3 events related to cutoffs in trial end points, such as time to decline of at least 6 points on the ALSFRS-R, and 8 exploring clinical events, including time to assisted ventilation, time to feeding tube, time to first hospitalization due to a serious adverse event [SAE], and time to first hospitalization due to ALS-related SAE).

Serum NfL analysis was prespecified to occur in the full analysis set (including placebo participants from other regimens), while plasma NfL was prespecified to occur in samples from this regimen only. For this regimen only, serum NfL was rerun with all samples from an individual on a single plate to reduce the impact of plate-to-plate variability on the analysis; for this reason, comparison with shared placebo was not conducted. These post hoc regimen-only results are presented.

Additional details about all measures and their analyses can be found in the statistical analysis plan (Supplement 2) and in the eMethods in Supplement 3.

Sample Size Calculation

Trial simulations were used to estimate sample size. A sample size of 160 per regimen with randomization 3:1 to CNM-Au8 (combined 60-mg and 30-mg dosages) or placebo and sharing of controls across concurrently enrolling regimens provided approximately 80% power to detect a 30% slowing in ALS progression (common across mortality and function) with a 1-sided type I error rate less than 2.5%.

Statistical Analysis

Analysis Populations

The primary analysis population, referred to as the full analysis set, included all participants randomized to the regimen active treatment group, all participants randomized to the placebo group within the regimen, and all placebo participants from other concurrent regimens (referred to as the shared placebo group). Additional prespecified efficacy analyses were performed using only regimen participants (CNM-Au8 and placebo), called the regimen-only analyses.

The population considered for safety analysis included participants who initiated treatment within regimen plus placebo participants from specified regimens who initiated treat-

ment in their respective regimen and were not known to be ineligible for this regimen. More details are available in the eMethods in Supplement 3.

Primary Analysis Methods

The primary analysis for the primary end point was a bayesian shared parameter model of function and survival that provided an integrated estimate of the increase or decrease in the rate of disease progression on treatment relative to control (denoted as the DRR).¹⁴ The DRR measured the ratio of ALSFRS-R slopes within the functional component or the hazard ratio (HR) within the survival component, with values less than 1 indicating slowing in disease progression. Detailed methods are available in the eMethods in Supplement 3.

Secondary Efficacy Analysis Methods

CAFS is a joint-rank analysis that compared the treatment and placebo groups through a series of pairwise comparisons between participants. In these pairwise comparisons, each participant in the treatment group was compared with each participant in the placebo group based on 2 key factors: survival time and change in ALSFRS-R score. Pairwise comparison scores for each participant were summed and then participants were ranked based on their score. The mean ranks for those in the treatment group and those in the placebo group were calculated and compared. Higher mean rank indicated less progression and longer survival for that group. A Wilcoxon rank-sum statistical test was used to compare the groups.

SVC was assessed using both a repeated measures linear mixed model and random slopes model with the following covariates centered to the analysis population: months since symptom onset, prebaseline slope of ALSFRS-R (delta-FRS), baseline riluzole and edaravone use, and their interaction with time.

Each participant's vital status was verified at the time of their last visit in the randomized clinical trial (RCT). Rates of death or death-equivalent events (PAV, defined as use of ventilation for >22 hours per day for more than 7 consecutive days) were analyzed using a Cox proportional hazards model, with covariates of prebaseline ALSFRS-R slope, months since symptom onset, riluzole and edaravone use at baseline, and age at baseline. Participants with no follow-up time or with an event at baseline were not included in the time-to-event models, but were included in the models with functional outcomes. Models handled missing data, including data missing due to death, as missing at random conditional on the observed data and model structure. Additional detailed methods are available in the eMethods in Supplement 3.

Exploratory Analyses

Information on prespecified exploratory and other supportive analyses, including subgroup analyses, can be found in the eMethods in Supplement 3.^{18,20,21}

Primary analyses compared the combined CNM-Au8 dosages with placebo, and prespecified analyses were also carried out in individual dosages. Analyses were performed using SAS version 9.4 (SAS Institute) and R version 4.1.2 (R Foundation). For the primary bayesian model, a threshold for significance

of 0.979 or greater for probability of DRR less than 1 was prespecified based on controlling type I error in simulations. If the primary analysis reached significance, then each secondary efficacy end point would be evaluated for significance in order: CAFS model, SVC, and survival using a comparison-wise criterion of 2-tailed $P < .05$. After the first of these end points failed to declare significance, no end points lower in the hierarchy could be declared significant. This sequential closed-testing procedure controlled the overall type I error rate at 5%. Nominal comparison-wise P values for secondary efficacy end points are also reported without adjustment for multiple comparisons. As with secondary analyses, models handled all missing data as missing at random, conditional on the observed data and model structure.

Results

Trial Participants

A total of 163 individuals were screened for the regimen. One participant died prior to the regimen screening visit and 1 timed out of the screening window and did not return, thus 161 were randomized to receive CNM-Au8 (30 mg [$n = 59$]; 60 mg [$n = 61$]) or regimen-specific placebo ($n = 41$) and an additional 123 participants were randomized to receive placebo on other concurrently enrolling regimens within the master protocol (Figure 1). All randomized participants within the regimen received their assigned drug, and all confirmed treatment initiation. The trial did not meet the cutoff for early futility and was completed as planned. In all, within the analysis population, 85% of participants in the placebo group and 92% of participants in the CNM-Au8 group completed the trial on the assigned drug (93% for the 30-mg dosage and 90% for the 60-mg dosage) (Figure 1). Five participants in the placebo group and 6 in the CNM-Au8 group (3 in the 30-mg dosage and 3 in the 60-mg dosage groups) discontinued the trial drug before the week 24 visit. One participant in the regimen placebo group died during participation in the RCT portion of the study, as did 1 in the 30-mg dosage group and 3 in the 60-mg dosage group.

Baseline demographic and disease characteristics are summarized in Table 1. Participants randomized within the CNM-Au8 regimen had a mean age of 57.7 years, 37.9% were female, and 98% were non-Hispanic or Latino White. Baseline characteristics were minimally different between groups, although most of the small differences that did exist would correlate with faster progression in the CNM-Au8 group. Considering the pooled CNM-Au8 and shared placebo groups, the pooled CNM-Au8 group had more participants with clinically definite ALS (48% vs 40%), fewer with clinically possible or clinically probable laboratory-supported ALS (16% vs 35%), and fewer participants in King's Stage 1 (11% vs 21%). Body mass index (calculated as weight in kilograms divided by height in meters squared) was higher in the regimen placebo group compared with the pooled CNM-Au8 group (28.4 vs 27.0, respectively). Median (IQR) plasma NfL was lower in the regimen placebo group at 75.7 (48.5-105.8) pg/mL than in the 30-mg CNM-Au8 group at 92.2 (59.7-120.6) pg/mL, but not

the 60-mg group at 75.0 (51.0-110.9) pg/mL. The mean baseline ALSFRS-R score was similar across randomization groups. Most participants (76%-80%) were taking riluzole and a minority (20%-26%) were taking edaravone at trial entry, with 19% to 23% taking both. Distribution of the ALSFRS-R score by visit and group is presented in eTable 1 in Supplement 3.

Primary Efficacy Outcome

The estimated DRR common to ALSFRS-R score and mortality was 0.97 (95% credible interval [CrI], 0.783-1.175; posterior probability of DRR < 1 , 0.65), suggesting no benefit or harm of CNM-Au8 at week 24 (Figure 2). When limiting the comparison to only the placebo participants within the CNM-Au8 regimen, there were similar results (DRR, 0.96 [95% CrI, 0.709-1.357]). The estimated mean slope of ALSFRS-R was -1.03 points per month (95% CrI, -1.176 to -0.892) in the shared placebo group and -1.00 points per month (95% CrI, -1.153 to -0.858) for the combined dosages of 30 mg and 60 mg daily of CNM-Au8 groups. The bayesian shared parameter model estimated that the mortality event rate was 0.007 events per month for the shared placebo group and 0.006 events per month for the combined CNM-Au8 dosage groups (Figure 2) (eTable 2 in Supplement 3).

Secondary Efficacy Outcomes

CAFS showed a shared placebo mean rank of 143.9 and combined CNM-Au8 dosage group mean rank of 140.5 ($P = .51$). Sensitivity analyses adjusting for baseline NfL did not change the inference ($P = .88$) (eMethods in Supplement 3). The estimate of the mean change in SVC over 24 weeks was -9.32 PPN per month in the combined CNM-Au8 group (30 mg, -7.84; 60 mg, -10.79) vs -8.53 PPN per month in the shared placebo group (difference, -0.78 PPN/month [95% CI, -4.25 to 2.68]) (eTable 3A and 3B in Supplement 3). Sensitivity analyses adjusting for baseline NfL did not change the inference (eMethods in Supplement 3); mean change in SVC over 24 weeks was -1.73 PPN per month in the combined CNM-Au8 group (95% CI, -2.18 to -1.29) vs -1.55 PPN per month in the shared placebo group (95% CI, -1.94 to -1.17).

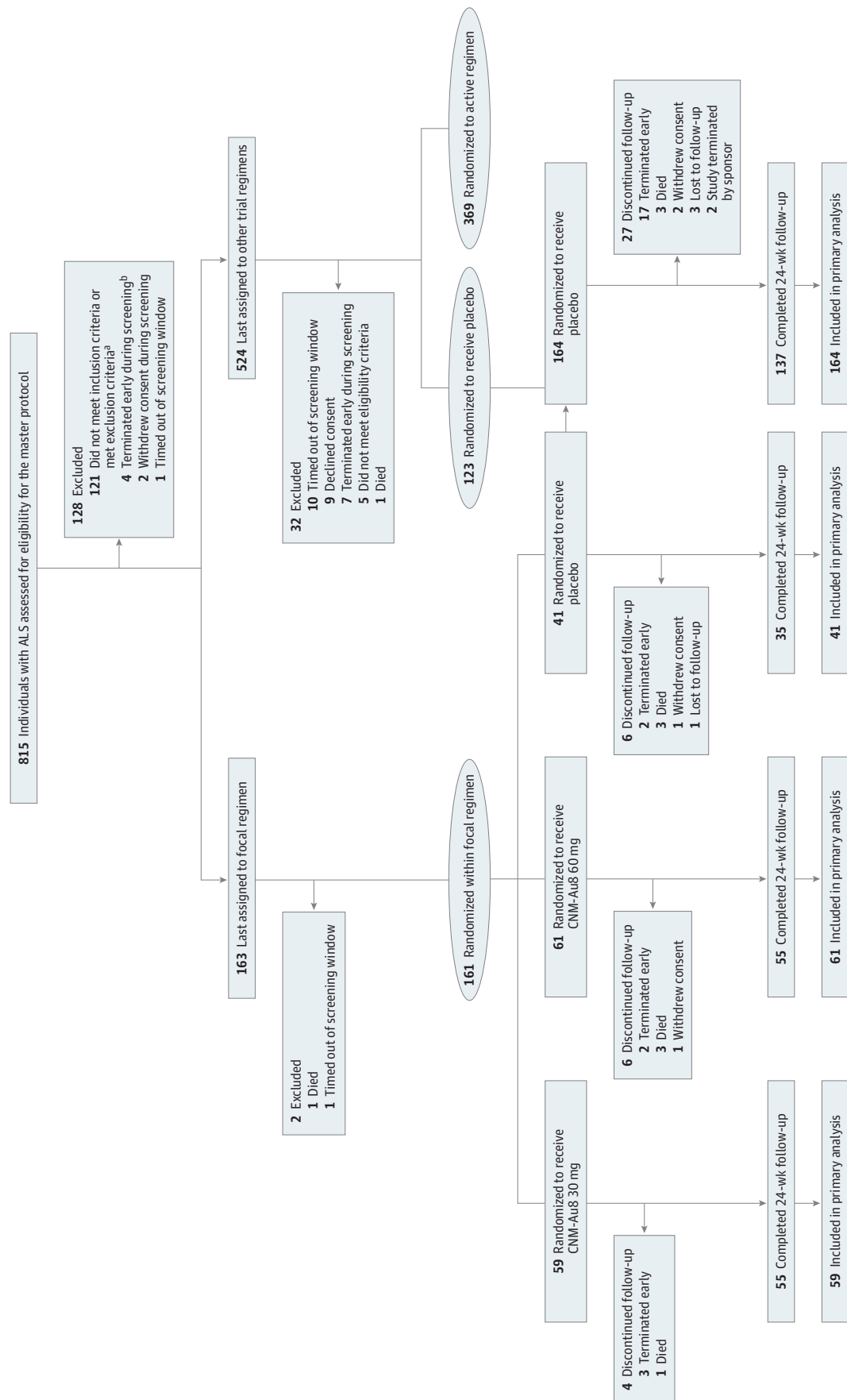
Proportions of death or PAV to week 24 were: 1 of 59 in the CNM-Au8 30-mg group, 4 of 61 in the CNM-Au8 60-mg group, and 9 of 162 in the shared placebo group. Of those, 4 of the 41 participants in the regimen placebo group and 2 in the shared placebo group were excluded due to having no follow-up time (1 had PAV at baseline and 1 terminated early at baseline visit). The cumulative adjusted HR for death or death equivalent in the combined CNM-Au8 dosage group compared with the shared placebo group was 0.46 (95% CI, 0.12-1.49) (Table 2; Figure 2B). Sensitivity analysis adjusting for baseline NfL did not change the inference (eMethods in Supplement 3); the HR for the combined CNM-Au8 dosage group vs the shared placebo group was 0.45 (95% CI, 0.11-1.50).

Exploratory Efficacy Outcomes

Survival Analyses

The cumulative adjusted HR for death or death equivalent in the 30-mg dosage group (1 event [1.7%]) was 0.06 (95% CI,

Figure 1. Recruitment, Randomization, and Follow-Up for the CNM-Au8 Regimen of the HEALEY ALS Platform Trial



^aParticipants could have multiple reasons for exclusion from the master protocol. The most common reasons were not meeting the criteria for slow vital capacity of 50% or greater (46%), having a clinically significant unstable medical condition other than ALS (31%), and using investigational treatments for ALS within 5 half-lives (if known) or 30 days (whichever was longer) prior to the screening visit (17%).

^bCommon reasons for termination before randomization were travel difficulties and enrolling in another study, whereas common reasons for termination after randomization were disease progression and perceived lack of efficacy. ALS indicates amyotrophic lateral sclerosis.

Table 1. Demographic and Baseline Characteristics of Participants in the HEALEY ALS Platform Trial Regimen C (CNM-Au8)

Characteristic	No. (%)			
	CNM-Au8		Placebo	
	30 mg (n = 59)	60 mg (n = 61)	Shared (n = 164) ^a	Regimen-specific (n = 41)
Sex				
Female	26 (44.1)	23 (37.7)	49 (29.9)	12 (29.3)
Male	33 (55.9)	38 (62.3)	115 (70.1)	29 (70.7)
Race				
Asian			2/160 (1.2) ^b	
Black or African American			6/160 (3.8) ^b	2 (4.9)
White	59 (100.0)	61 (100.0)	151/160 (94.4) ^b	38 (92.7)
Multiple races			1/160 (0.6) ^b	1 (2.4)
Ethnicity				
Not Hispanic or Latino	56 (94.9)	60/60 (100.0)	157/163 (96.3)	38 (92.7)
Age, mean (SD), y	57.7 (10.2)	58.6 (9.9)	57.2 (11.3)	57.0 (11.7)
BMI, mean (SD)	27.4 (5.3)	26.6 (4.8)	27.3 (5.0)	28.4 (5.5)
Bulbar onset	10 (16.9)	8 (13.1)	29 (17.7)	6 (14.6)
El Escorial criteria ^c				
Clinically definite ALS	28 (47.5)	30 (49.2)	66 (40.2)	14 (34.1)
Clinically probable ALS	22 (37.3)	21 (34.4)	40 (24.4)	10 (24.4)
Clinically probable ALS – laboratory-supported	8 (13.6)	8 (13.1)	42 (25.6)	16 (39.0)
Clinically possible ALS	1 (1.7)	2 (3.3)	16 (9.8)	1 (2.4)
King's stage ^d				
1 (1 clinical region involved)	3 (5.1)	10 (16.4)	34 (20.7)	9 (22.0)
2 (2 clinical regions involved)	18 (30.5)	18 (29.5)	39 (23.8)	11 (26.8)
3 (3 clinical regions involved)	22 (37.3)	19 (31.1)	45 (27.4)	7 (17.1)
4a/4b (nutritional failure)	1 (1.7)		1 (0.6)	
4b (respiratory failure)	15 (25.4)	14 (23.0)	45 (27.4)	14 (34.1)
Riluzole and edaravone use at baseline				
Riluzole use	45 (76.3)	49 (80.3)	126 (76.8)	32 (78.0)
Edaravone use	12 (20.3)	16 (26.2)	41 (25.0)	10 (24.4)
Months since ALS symptom onset, mean (SD)	21.2 (8.6)	24.2 (8.5)	21.9 (8.7)	21.9 (8.5)
Months since ALS diagnosis, mean (SD)	9.8 (5.2)	11.1 (6.6)	10.3 (6.1)	10.0 (5.6)
Slow vital capacity, PPN, mean (SD)	74.4 (16.0)	76.0 (16.3)	76.0 (16.5)	76.1 (16.8)
ALSFRS-R score, mean (SD)				
Total score ^e	34.5 (5.8)	34.0 (7.3)	35.1 (6.7)	36.1 (5.91)
Bulbar score	9.8 (2.1)	10.0 (2.5)	10.0 (2.3)	10.6 (2.1)
Fine motor score	7.3 (2.7)	6.9 (3.4)	7.6 (3.1)	7.8 (3.4)
Gross motor score	7.2 (2.8)	6.7 (3.2)	7.3 (3.1)	7.3 (2.8)
Combined motor score	14.4 (4.9)	13.6 (5.7)	14.9 (5.4)	15.1 (5.4)
Respiratory score	10.3 (2.4)	10.5 (2.0)	10.2 (2.1)	10.3 (2.0)
Prebaseline ALSFRS-R slope (points/month), mean (SD)	0.8 (0.6)	0.7 (0.5)	0.7 (0.4)	0.6 (0.4)
Serum NfL, median (IQR), pg/mL	67.6 (44.3-91.6)	62.5 (44.4-95.1)	NA	51.6 (33.0-76.4)
Plasma NfL, median (IQR), pg/mL	92.2 (59.7-120.6)	75.0 (51.0-110.9)	NA	75.7 (48.5-105.8)

Abbreviations: ALS, amyotrophic lateral sclerosis; ALSFRS-R, Revised Amyotrophic Lateral Sclerosis Functional Rating Scale; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); LMN, lower motor neuron degeneration; NA, not applicable; NfL, neurofilament light; PPN, percent predicted normal; UMN, upper motor neuron degeneration.

^a The shared placebo group includes regimen-specific placebo and shared placebos from other regimens.

^b Race was unknown or unreported for 3 participants and ethnicity for 1 participant.

^c Clinically definite ALS is defined by the presence of UMN and LMN signs in 3 regions of the central nervous system, such as brainstem, cervical, thoracic, or lumbar; clinically probable ALS, UMN and LMN signs in at least 2 regions, with some UMN signs rostral to LMN signs; clinically probable ALS – laboratory-supported, clinical signs of UMN and LMN dysfunction in only 1 region or UMN signs alone are present in 1 region, and LMN signs defined by electromyography criteria are present in at least 2 limbs, with application of neuroimaging and clinical laboratory protocols to exclude other causes; clinically possible ALS, clinical signs of UMN and LMN dysfunction found together in 1 region, UMN signs found alone in 2 or more regions, or LMN signs found rostral to UMN signs and diagnosis of clinically probable – laboratory-supported ALS cannot be proven in conjunction with electrodiagnostic, neurophysiologic, neuroimaging, or clinical laboratory studies. Other diagnoses must have been excluded to accept a diagnosis of clinically possible ALS.

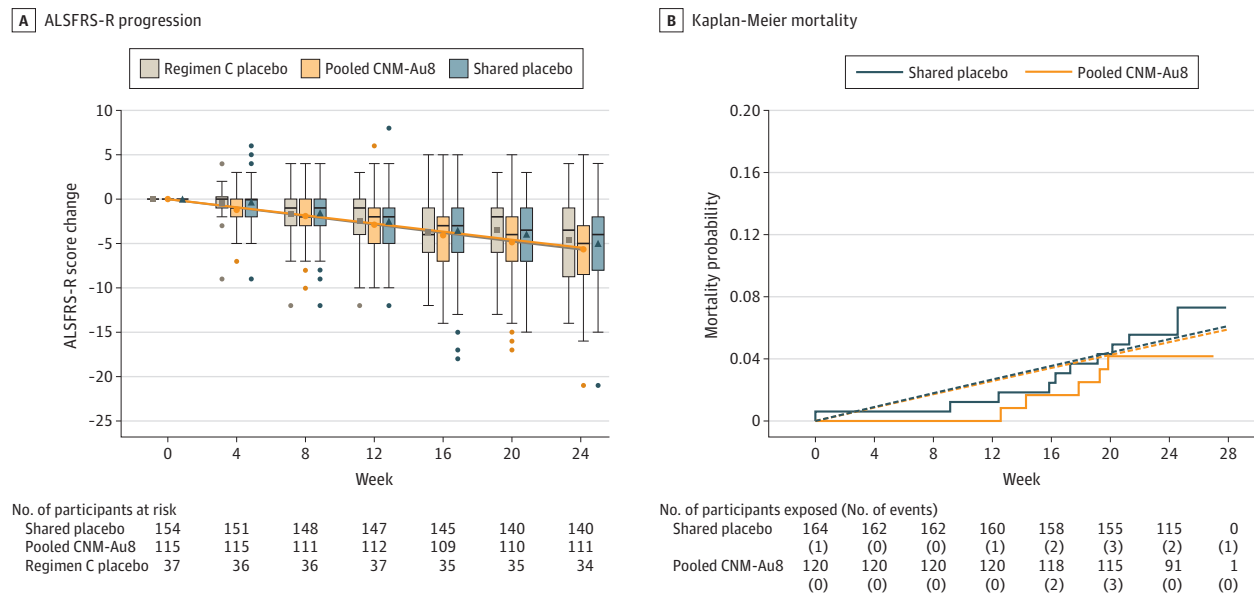
^d The King's Staging System has 5 stages, with stage 1 involving 1 clinical region; stage 2, a second clinical region; stage 3, a third clinical region; stage 4, nutritional or respiratory failure; and stage 5, death.

^e Scores range from 0 to 48, with higher scores indicating better function.

0.002 -0.56) and in the 60-mg dosage group (4 events [6.6%]) was 0.96 (95% CI, 0.25-3.08) compared with the shared placebo group (9 events [5.6%]) (Table 2; Figure 2B). The combined CNM-Au8 dosage group (5 events [4.2%]) relative to the

shared placebo group (9 events [5.6%]) adjusted HR was 0.46 (95% CI, 0.12-1.49) (Table 2).

For within-regimen analyses, the 30-mg CNM-Au8 dosage group (1 event [1.7%]) adjusted HR was 0.03 (95% CI,

Figure 2. Change in Disease Severity Over 24 Weeks Measured by the Revised Amyotrophic Lateral Sclerosis Functional Rating Scale (ALSFRS-R) and Survival

A, The bottom of each box represents the 25th percentile of the ALSFRS-R change from baseline values and the top of the box represents the 75th percentile. The horizontal line within each box represents the median value. The whiskers extend to the most extreme data point, which is no further than 1.5 times the length of the box away from the box, with outlier values beyond this range represented as solid circles. Points are the raw mean values at each visit. The solid lines are the model-estimated change in ALSFRS-R over time, adjusting for covariates. The estimate for the regimen placebo shares

information from placebo participants in other regimens. The numbers below the panel show the number of participants with known ALSFRS-R outcomes at each visit. The summary excludes all ALSFRS-R data from participants who died or had a death-equivalent event (permanent assisted ventilation [PAV]). B, Kaplan-Meier curves for death and PAV per group (solid lines) and the model-estimated exponential curves (dashed lines) for the active treatment and pooled placebo. The numbers below the panel indicate the number of participants exposed and the number of death and PAV events per group.

Table 2. Survival Analyses

PAV-free survival analysis comparison groups	Observed proportions		Adjusted Cox regression, ^b HR (95% CI) ^c	P value
	CNM-Au8	Placebo ^a		
CNM-Au8 pooled vs shared placebo	5/120	9/162	0.46 (0.12-1.49)	.22
CNM-Au8 30 mg vs shared placebo	1/59	9/162	0.06 (0.002-0.56)	.04
CNM-Au8 60 mg vs shared placebo	4/61	9/162	0.96 (0.25-3.08)	.95
CNM-Au8 pooled vs regimen C placebo	5/120	4/41	0.25 (0.05-1.09)	.06
CNM-Au8 30 mg vs regimen C placebo	1/59	4/41	0.03 (0.0004-0.36)	.03
CNM-Au8 60 mg vs regimen C placebo	4/61	4/41	0.47 (0.10-2.16)	.32

Abbreviations: HR, hazard ratio; PAV, permanent assisted ventilation.

^a Two shared placebo participants were not included due to having no follow-up time; 1 had PAV at baseline and the other terminated early at baseline.

^b Models adjusted for age, sex, time since symptom onset, prebaseline ALSFRS-R slope, riluzole use at baseline, and edaravone use at baseline.

^c CIs were calculated from the profile log-likelihood.

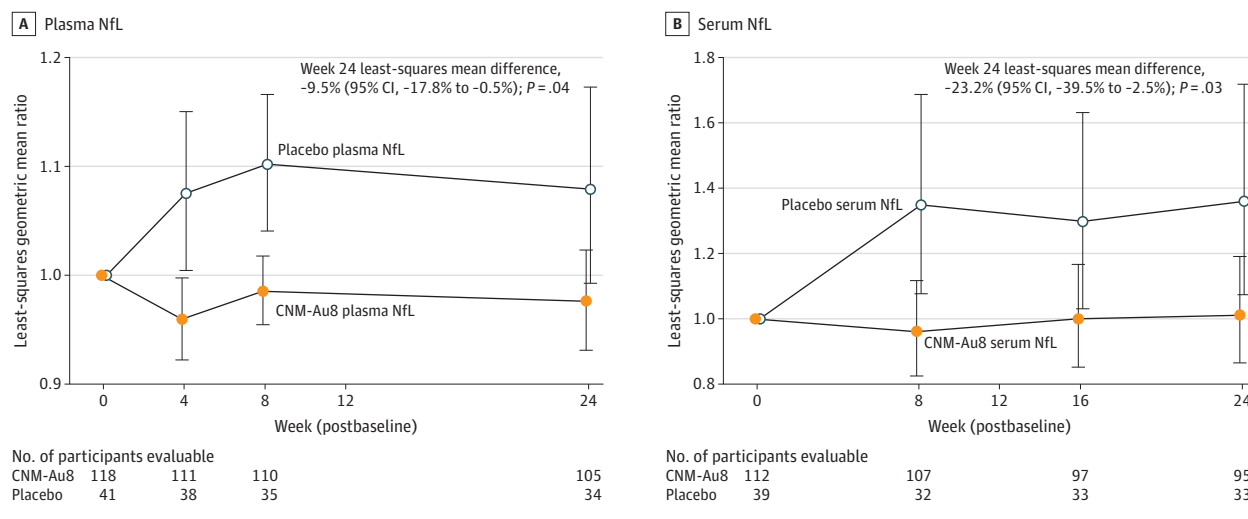
0.0004-0.36) and the 60-mg dosage group (4 events [6.6%]) adjusted HR was 0.47 (95% CI, 0.10-2.16) compared with the regimen placebo group (4 events [9.8%]) (Table 2). The combined CNM-Au8 dosage group (5 events [4.2%]) relative to regimen placebo (4 events [9.8%]) comparators adjusted HR was 0.25 (95% CI, 0.05-1.09) (Table 2). Additional efficacy analyses did not show statistically significant differences among other analysis populations (Table 2).

Other Time-to-Event Analyses

Exploratory time-to-event analysis compared combined CNM-Au8 therapy with shared and within-regimen placebo, evalu-

ating the time to key ALS morbidity milestones adjusting for baseline covariates. In the regimen-specific placebo group, this demonstrated (1) assisted ventilation for 16 participants (18%) in the active group vs 9 (31%) in the placebo group (HR, 0.40 [95% CI, 0.17-1.01]), (2) gastrostomy tubes in 14 participants (12%) in the active group vs 7 (17%) in the placebo group (HR, 0.37 [95% CI, 0.14-1.04]), (3) first hospitalization due to ALS-related SAE in 5 participants (4%) in the active group vs 3 (7%) in the placebo group (HR, 0.23 [95% CI, 0.04-1.33]), and (4) first hospitalization due to SAE in 14 participants (12%) in the active group vs 7 (18%) in the placebo group (HR, 0.48 [95% CI, 0.18-1.33]).

Figure 3. Serum and Plasma Neurofilament Light (NfL)



NfL was tested in plasma and serum samples longitudinally at baseline, week 4 (plasma only), week 8, week 16, and week 24. Values were converted to natural log (Ln) scale prior to analysis and the difference in least-squares means is shown at each time point. After back-transformation to compare values in the placebo and combined dosage CNM-Au8 treatment groups, in both plasma and

serum, the NfL levels were increased in the regimen placebo group (7.9% in plasma and 30.8% in serum) and were decreased or essentially unchanged in the CNM-Au8 treatment group (-2.3% in plasma and 0.4% in serum). The differences in change comparing the placebo and CNM-Au8 treatment groups were nominally significant for both plasma ($P = .04$) and serum ($P = .03$).

Subgroup Analyses

Subgroup analyses did not reveal any predefined subgroups with significant differences in the rate of ALSFRS-R progression comparing the combined CNM-Au8 dosages with shared placebo (eTable 4 in Supplement 3).

Biomarker Analyses

Serum NfL levels showed a 30.8% geometric mean increase in the regimen placebo group (43.1 pg/mL at baseline vs 56.5 pg/mL at 24 weeks) compared with a change of 0.4% in the combined CNM-Au8 dosage groups (60.6 pg/mL at baseline vs 60.8 pg/mL at 24 weeks; treatment difference, -23.2% geometric mean ratio [95% CI, -39.5% to -2.5%]; $P = .03$) (Figure 3). To assess whether this serum NfL result was sensitive to outliers, a post hoc analysis was performed that excluded observations with very low NfL values (<4 pg/mL or <2% of the maximum for a given participant). This excluded 1 baseline NfL for 1 participant in the placebo group. In this sensitivity analysis, the geometric mean change of NfL for the combined CNM-Au8 group was 0.8% and 11.6% for the placebo group, a difference of -9.7% (95% CI, -18.5% to 0.1%; $P = .05$) (eFigure in Supplement 3). This result suggests that the magnitude but not directionality of the analysis of serum NfL was sensitive to the effect of an outlier. Plasma NfL showed an increase of 7.9% in the geometric mean of the regimen placebo group (72.8 pg/mL at baseline vs 78.5 pg/mL at 24 weeks) compared with a geometric mean decrease of 2.3% in the combined CNM-Au8 dosage groups (80.3 pg/mL at baseline vs 78.5 pg/mL at 24 weeks; treatment difference, -9.5% [geometric mean ratio] [95% CI, -17.8 to -0.4%]; $P = .04$) (Figure 3) (eTable 3A and 3B in Supplement 3).

No other differences in exploratory outcomes suggested benefit for CNM-Au8 in the combined or individual dosages relative to placebo (eTable 3A and 3B in Supplement 3).

Safety and Tolerability

In the safety analysis population, which included 120 participants who received CNM-Au8 and 163 participants across the shared placebo groups, nearly all participants (combined dosages of CNM-Au8, 93%; shared placebo, 90%) reported 1 or more treatment-emergent adverse event during the trial; few led to interruption of trial drug dosing (eTable 5 in Supplement 3).

Adverse events that were at least 5% more common in the combined CNM-Au8 group compared with the shared placebo group included diarrhea (19% vs 7%) and nausea (14.2% vs 8.6%). Adverse events that were at least 5% less common in the CNM-Au8 group included fatigue (10.8% vs 18.4%) and muscular weakness (20% vs 27.6%) (eTable 5 in Supplement 3).

Serious treatment-emergent adverse events occurred in 9% of the shared placebo group, 10% of the 30-mg dosage group, 16% of the 60-mg dosage group, and 17% of the regimen placebo group ($P = .31$) (eTable 5 in Supplement 3). No SAEs and no deaths were considered related to the trial drug.

Discussion

This 24-week RCT did not demonstrate a slowing of ALS progression with CNM-Au8 treatment as a combined dosage group or in the individual 30-mg or 60-mg dosage groups compared

with placebo. No benefit of CNM-Au8 on key secondary end points was seen at 24 weeks.

Survival analyses, adjusted for baseline parameters, are hypothesis generating. A prespecified exploratory analysis of the 30-mg group, but not the 60-mg group, compared with the shared and regimen placebo groups showed a nominal survival benefit after adjusting for baseline covariate differences, albeit with small numbers of events (a total of 13 events in the RCT period in the shared placebo group and the combined CNM-Au8 groups).

Prespecified exploratory time-to-event analyses revealed point estimates in favor of the CNM-Au8 combined group compared with the regimen placebo group for time to placement of a gastrostomy tube, initiation of noninvasive ventilation, and hospitalization related to SAEs, although these differences were not statistically significant. These findings also must be interpreted cautiously, given the large number of exploratory outcomes assessed in this study without control for multiple comparisons.

Analysis of serum and plasma NfL showed an increase in the regimen placebo group consistent with some prior reports of observational cohorts.¹⁹ The combined CNM-Au8 group did not show an increase, thus there was a nominally significant difference in the 24-week NfL change between groups. Data from a 2022 trial suggest that treatment resulting in substantial reductions in NfL can predict clinical benefit in people with ALS.²² Smaller changes may be important, although the degree of change in NfL necessary to predict clinical benefit is unknown.

In terms of safety and tolerability, AEs were common, but few led to treatment discontinuation. There were no SAEs and no deaths that were considered related to the trial drug.

Limitations

This study has limitations. First, this was a 24-week study, and the elimination half-life of CNM-Au8 is 28 days, long enough that a steady state would have only been achieved in the latter part of the RCT. Longer-term evaluation of efficacy and safety of CNM-Au8 as a part of the ongoing open-label extension and expanded access programs will provide additional data for analysis. Second, although the baseline demographics were largely balanced, there were small baseline trends toward slightly more advanced or faster-progressing ALS disease among participants in the CNM-Au8 groups. Correction for baseline differences in the analytic models was aimed at minimizing the effect of these chance differences. Third, the mixed model analyses of secondary outcomes assumed data are missing at random, including death, and should be taken into consideration in the interpretation of results. However, with few deaths seen in the study, any bias due to the missing at random assumption is likely minimal.

Conclusion

CNM-Au8 did not improve the primary efficacy outcome of ALS disease progression measured in this 24-week RCT.

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